Experimental conjecture: On representations of Gaussian integers by primes and powers of $1+i$

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Conjecture. Let $z = a + bi \in \mathbb{Z}[i]$ with $a + b$ odd. Then z admits a representation of the form

$$z \;=\; \pi \;+\; \varepsilon\,(1+i)^k,$$

where π is a Gaussian prime, $\varepsilon \in \{\pm 1, \pm i\}$ is a unit, and $k \geq 0$ is an integer. No counterexamples were found within the tested range ($a^2 + b^2 < 10^{11}$).

Historical remark. The classical one-dimensional analogue concerns representations of odd integers in the form

$$n = p + 2^k.$$

This problem goes back to de Polignac, who asked in 1849 whether every odd integer greater than 3 can be written in this way [1]. Although this is false, Romanoff proved in 1934 that the set of integers representable as a prime plus a power of 2 has positive lower density [3] (see [5] for a generic field case and [6] for the case of Gaussian integers).

Possible route to a counterexample. As in [2, 4] we could try to construct for each unit $\varepsilon_j \in \{1, -1, i, -i\}$, a covering system of moduli $\{m_{ij}\}_{j=1}^4$ together with pairwise disjoint sets of distinct Gaussian primes $\{p_{ij}\}_{j=1}^4$, such that

$$(1+i)^{m_{ij}} \equiv 1 \pmod{p_{ij}}$$

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for all relevant indices. One would then attempt to combine these congruences to force $\pi = z - \varepsilon_j(1+i)^k$ to be divisible by a Gaussian prime from the corresponding set $\{p_{ij}\}$, for each $j = 1, 2, 3, 4$, depending on the residue class of k. No such covering systems satisfying the required disjointness and congruence conditions were yet found.

References

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